

Topological Quantum Computing

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Introduction

0.1 Course Info

0.1.1 Teachers

Konstantin Wernli – kwernli@imada.sdu.dk

William Mistegard – wem@imada.sdu.dk

0.1.2 Exercise Classes

Once a week with at least 3 solutions handed in.

0.1.3 Exam

1. Assessment is based on four graded exercise sheets, and
2. Final oral exam consisting of:
 - 10–12 minutes presentation,
 - 13–15 minutes discussion and questions,
 - 4–5 possible topics, with 30 minutes of preparation time.
3. Tentative exam date: January 18, 2027

0.1.4 Books

N/A

1 First Lecture

1.1 Topological Quantum Computing

Different fields:

- Condensed matter physics: topological phases of matter
- Topological Quantum Field Theory
- TQFT's in math
- Anyon models/UMTC in math

1.2 What is Topological Quantum Computing?

Digital vs analog quantum computing.

Problem They are prone to errors and noise, such as spam error during state preparation, gate fidelities as well as decoherence.

The idea of TQC is to use very special physical systems where information is encoded in discrete global properties. These are topological phases of matter.

Certain phases in $2+1$ dimensions, with 2 spatial dimensions and 1 time, admit anyons. Anyons are a special class of (quasi) particles with statistics different from bosons and fermions.

In more than $2+1$ dimensions $T^2 = 1 \rightarrow T = \pm 1$ $+1 = e^{i2\pi i}$ bosons $-1 = e^{i\pi i}$ fermions

In $2+1$ dimension $T_2 \neq 1$, meaning it can get any phase $T = e^{i\Theta}$

If $T = e^{i\Theta} \rightarrow$ abelian anyons, but there are non-abelian anyons where T can be a unitary acting on $\mathcal{H}$. These are the unitary we want to use in TQC.

1.3 Non-abelian anyons

Described by $|\psi\rangle \in \mathcal{H}$ when exchanging anyons

$$|\psi\rangle \rightarrow U |\psi\rangle,$$

for some unitary U .

Braids form a group.

Companies trying to develop non-abelian anyons: Microsoft and Nokia Simulate anyons on quantum computer (topological codes): Google and Quantinuum

1.4 Plan

Math foundations:

- Groups (braid),
- Representation,
- Category theory,
- Topology.

TQC:

- Anyon models,
- Examples,
- Equivalence to digital quantum computing,
- Relation to TQFT's, etc.

TQC in practice:

- Topological phases of matter,
- Experimental realizations.

Topological codes:

- Surface codes,
- Other classes of topological codes.

2 Second Lecture

2.1 Mathematical Foundations

A group is a set G with a binary operation $G \times G \rightarrow G$, $(g, h) \mapsto g \cdot h = gh$ such that it is associative for all $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ there is $e \in G$ such that for all $a \in G$ $e \cdot a = a \cdot e = a$, and that for all $a \in G$ is $a^{-1} \in G$ such that $aa^{-1} = a^{-1}a = e$
In general $gh \neq hg$. Groups which have this property are called commutable or abelian.

Examples

- $(\mathbb{Z}, +)$: yes
- $(\mathbb{N}, +)$: no
- $(\mathbb{Z}, \cdot)$: no
- $(\mathbb{Q}, \circ)$: no

These are all commutative infinite examples.

2.1.1 Finite Groups

$\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}n = \{0, \dots, n-1\}$ with abelian *mod* n

$\mathbb{Z}_2 = \{0, 1\}$

+	0	1
0	0	1
1	1	0

$\mathbb{Z}_3$

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

2.1.2 Symmetric Group

S_n =: permutations of $\{1, \dots, n\} = [n] = \{\sigma, [n] \rightarrow [n] | \text{bijective}\}$

$S_1 = \{e\}$

$S_2 = \{e, (12)\}$

o	e	(12)
e	e	(12)
(12)	(12)	e

$S_3 = 3! = 6 = \{e, (123), (132), (23), (12), (13)\}$ is a non-abelian group
 S_3 needs at least 2 generators: $(12), (23)$ for example.

0	e	12	13	23	123	132
e	e	12	13	23	123	132
12	12	e		132		
13	13		e			
23	23	123		e		
123	123				132	e
132	132				e	123

2.2 Generators and Relations

We say a group G is generated by $g_1 \rightarrow g_n \in G$ if all elements in G are products of $g_1 \rightarrow g_n$ and their inverses.

A relation is a combination of generators.

$(\mathbb{Z}, +)$ is generated by 1.

$(\mathbb{Z}_n, +)$ is also generated by 1.

$\mathbb{Z}_3$ is generated by 1, 2.

$\mathbb{Z}_n$ is generated by all m with $\gcd(m, n) = 1$.

Relations in $S_n : \sigma_i^2 = e$

$\sigma_i \sigma_j = \sigma_j \sigma_i$

$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \rightarrow (12)(23)(12) = (23)(12)(23)$

Braid group on n strands generators $\sigma, \dots, \sigma_{n-1}$ but not $\sigma^2 = e$

2.3 Representation

Definitions A group homomorphism is a map $\Phi : G \rightarrow H$ such that $\Phi(g \cdot h) = \Phi(g) \cdot \Phi(h)$

Example $(\mathbb{Z}_2, + \rightarrow (S_2, \circ))$

Definition A group isomorphism is a group homomorphism Φ such that Φ is bijective and Φ^{-1} is a group homomorphism.

2.3.1 Linear Groups

V vector space $GL(V) = \{f : V \rightarrow V | f \text{ linear, invertable}\}$, $GL(\mathbb{R}^n) = GL_n(\mathbb{R}) = \{n \times n \text{ matrices}\}$.

We say (V, e) is a representation of G if there is a group homomorphism $\rho : G \rightarrow GL(V)$

Anyons live in representations of the braid group.

For the braid group, any $Z \in \mathbb{C}^\times$